

YAWER NAZIR RATHER

Computer Science & Engineering | Cybersecurity · Artificial Intelligence
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ABOUT ME

Computer Science and Engineering graduate interested in cybersecurity, artificial intelligence and quantum cryptography. Experience spans machine learning research, secure systems, software development and embedded applications, with a focus on applying computational methods to practical problems and building reliable, intelligent technologies.

EDUCATION

Bachelor of Technology in Computer Science & Engineering 2022 - 2026
Central University of Kashmir, Srinagar, India

- **Percentage: 74%.**
- **Relevant coursework:** Computer Programming, Object Oriented Programming, Data Structures, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Computer Networks, Network Security, Artificial Intelligence, Machine Learning, Deep Learning, Discrete Structures, and Mathematics (Differential Equations, Probability and Statistics).
- **Programming languages:** C, C++, Java and Python.

Higher Secondary School Examination (Class XII - Science/PCM) Completed 2020
Space Age Higher Secondary School, Srinagar, India

- Percentage: 82% | Grade: 8.6

Secondary School Examination (Class X) Completed 2018
New Generation Public School, Srinagar, India

- CGPA: 9.8 / 10.0

RESEARCH EXPERIENCE

Research Intern - Deep Learning & Biometrics Dec 2025 - Present
Indian Institute of Technology Jammu, India | Supervisor: Dr. Karan Nathwani

- Developed deep learning models for multimodal classification using the MM-AAD dataset, with an emphasis on representation learning and adaptation across recording conditions.
- Built a Cross-Stimulus Autoencoder (CSAE) to align audiovisual EEG representations with the feature distribution expected by a frozen, pretrained DARNet classifier, enabling adaptation without retraining the classifier.
- Trained and evaluated models in PyTorch across 30 labelled participants; compared random-window and trial-tail partitions, with identity initialization and control experiments to assess the learned representation mapping.
- Worked with iris and voice datasets for multimodal biometrics.

PUBLICATIONS & RESEARCH MANUSCRIPTS

Accepted Conference Paper

- Yash Paul, Shahid Sultan, Nazima Yousuf, **Yawer Nazir Rather**, and Wikas Khursheed. "An Efficient Graph-Mamba Network for EEG Auditory Attention Decoding." **Accepted**, IEEE IC2E3 2026, NIT Uttarakhand.

Co-authored a deep learning architecture combining hypergraph neural networks, Mamba state-space models and contrastive representation learning for classification across subjects.

Submitted Manuscript

- Rantu Buragohain, **Yawer Nazir Rather**, Sunil Kumar Kopparapu, and Karan Nathwani. "Cross-Stimulus EEG Representation Alignment for Auditory Attention Decoding Across Sessions." **Submitted to ICASSP 2027.**

CERTIFICATIONS & HONORS

- **Certificate of Appreciation - Session Moderator, 2AI 2026:** International Conference on Applied Artificial Intelligence, Central University of Kashmir, June 2026.
- **Building RAG Agents with LLMs - NVIDIA,** December 2025.
- **Certified in Cybersecurity (CC) - ISC2.**
- **Cybersecurity Fundamentals - HarvardX (edX).**
- **Arduino Timers & Interrupts - Udemy.**

PROJECTS

Beyond Signatures: Semantic Intelligence Against Generative Malware Evolution

2026

B.Tech Final-Year Project | Cross-Modal Transformer with Multi-Head Latent Queries (CMT-MHLQ)

- Developed an AI-based malware detector using a Cross-Modal Transformer with Multi-Head Latent Queries to identify malicious behaviour despite source-code rewriting and LLM-generated variants.
- Fused system-call sequences, process-resource graphs, process-lineage graphs and statistical profiles with Transformer/graph encoders and latent-query cross-attention; collected behaviour using Xen, DRAKVUF, LibVMI and AltP2M.
- Implemented generation-balanced training, orthogonality regularisation and collector-invariance learning; evaluated across five seeds and five folds (25 checkpoints), with family-disjoint splits and parent/sibling leakage controls.
- Achieved **0.9979 ± 0.0006 ROC-AUC** and **97.25% true-positive rate at a 1% calibration false-positive rate** on standard detection; retained **96.98% ± 10.50% recall** on source-rewritten L4 variants and **98.81% ± 0.73%** on cross-language LLM-generated L5 variants.
- Conducted held-out-family zero-day evaluation, modality ablations, collector-bias analysis and behavioural-integrity checks to assess robustness and generalization.

Cloud-Native Malware Analysis Lab

Oct 2025

- Deployed a Docker-based malware sandbox on Google Cloud Platform; automated sample execution, network-traffic capture and indicator-of-compromise extraction, with ELK Stack visualizations.

Kernel Watchtower: eBPF-Based Rootkit Detection

May - Jun 2025

- Built an eBPF/BCC monitoring tool to trace suspicious system calls and process execution, with a Python userspace agent for alert logging and threat visualization.

Smart Parking Management System

Aug - 2024

- Built an Arduino Uno prototype in C++ using an HC-SR04 ultrasonic sensor to detect parking-space occupancy, an RGB status indicator and a 16×2 I2C LCD for live monitoring.
- Implemented three-reading validation to reduce false occupancy changes from transient sensor noise, with visual guidance for available and occupied spaces.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, Java, SQL, Bash, PHP and 8086 Assembly.
- **Machine learning:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, PyTorch Geometric, graph neural networks, MNE-Python and OpenCV.
- **NLP and LLMs:** Retrieval-augmented generation, BERT, semantic search and vector databases.
- **Security and systems:** Xen, DRAKVUF, LibVMI, eBPF/BCC, Wireshark, Ghidra, Metasploit, Linux, Docker, Kubernetes, Git, GCP and AWS.
- **Methods and embedded systems:** Representation learning, multimodal fusion, statistical evaluation, optimization, Arduino and embedded C++.

LEADERSHIP & ACTIVITIES

- **Institute Ambassador, DEFCON Srinagar:** Organized capture-the-flag competitions and security workshops for 100+ students; represented Central University of Kashmir.
- **Head, University Research Club:** Coordinated research workshops and seminars; mentored peers in research methods and academic writing.
- **Event Coordinator, Cyber Conclave 2025:** Managed logistics and speaker sessions for the university cybersecurity summit.
- **National Service Scheme:** Led Swachh Bharat Abhiyan activities and coordinated rural blood donation camps.

LANGUAGES

- **English:** C1 (listening, reading, writing and speaking). **Urdu and Kashmiri:** Native / bilingual proficiency.